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MILITARY SPECIFICATION

HOSE ASSEMBLY, RUBBER, HYDRAULIC AND PNEUMATIC, JETTING (400 PSI WORKING PRESSURE)

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for one type of hose assembly for hydraulic and pneumatic jetting in pile driving, rock drilling, and air compressor service.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-775 - Hose, Hose Assemblies; Rubber, Plastic, Fabric or Metal (Including Tubing); and Fittings, Nozzles, and Strainers.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-No. 162 - Hose, Rubber, Visual Inspection Guide for.
FED-STD-No. 601 - Rubber, Sampling and Testing.

MILTARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D380 - Testing Rubber Hose.
D412 - Rubber Properties in Tension.
D413 - Rubber Property Adhesion to Flexible Substrate.
D471 - Rubber Property Effect of Liquids.
D518 - Rubber Deterioration - Surface Cracking.
D573 - Rubber Deterioration in Air Oven.
D1149 - Rubber Deterioration - Surface Ozone Cracking in a Chamber (Flat Specimen).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a complete hose assembly for the first article inspection and approval (see 4.2.1 and 6.5).

3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.2.1 Inner tube. The rubber compound used for the inner tube shall be compounded from synthetic rubber to resist water and oil. The inner tube shall have a smooth bore, free of pitting, dirt, and mandrel lubricants as consistent with good manufacturing practice, and shall have a minimum thickness of 0.093 inches.

3.2.2 Reinforcement. The reinforcement shall consist of sufficient plies of wrapped duck, spiral or braided yarns of cotton or synthetic fibers, or wire, or a combination of these necessary to meet the requirements of this specification and be suitable for the purpose intended. Where used, the layers of duck will be of cotton or synthetic fibers free from mechanical defects and applied on a bias of 45° with edges lapped at least 0.5 inch.

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3.2.3 Cover. The cover shall consist of an oil-, ozone-, and abrasion-resistant compound made from polymerized chloroprene or an equally oil-, ozone-, and abrasion-resisting material to meet the requirements of this specification. The cover shall have a minimum thickness of 0.063 inch.

3.3 Construction. Hose assemblies shall consist of tube, reinforcement and cover as described in 3.2.1, 3.2.2, and 3.2.3 with the fabric, yarn, or wire plies suitably impregnated with rubber. They shall be fitted with swivel connections as described in 3.4.

3.4 Hose end fittings. The ends of the hose shall be fitted with pearlitic malleable iron, interlocking-type female couplings. Couplings shall be provided with a wing nut with wrenching or knock lug features for quick loosening. The wing nut shall have NH hose threads to hold the steel adapter spud against the ground seat. The spud shall have internal NPT threads on one end and external NH threads with a mating ground joint seat on the other end. A 4-bolt clamp with four fingers shall be used to hold the shank to the hose. Threads shall be in accordance with FED STD H28.

3.5 Physical requirements. The hose and hose assembly shall conform to the requirements specified in Table I and the following paragraphs when tested as specified in Section 4.6.

TABLE I. Performance requirements.

Hydrostatic test pressures (minimum)	
Proof pressure (psi)	800
Burst pressure (psi)	1600
Tensile strength of rubber compound	
(before aging)(minimum)	
Tube, psi	850
Cover, psi	1000
Elongation (minimum)	
Tube (percent)	175
Cover (percent)	200
Adhesion (minimum)	
Between plies and between rubber parts (pounds)	12

3.5.1 Resistance to water.

3.5.1.1 Swelling. The increase in the volume of the tube shall not exceed 20 percent when tested as described in 4.6.4.

3.5.2 Resistance to oil. The tensile strength of the tube and cover shall not be less than 700 pounds per square inch (psi) when tested as specified in 4.6.7.

3.5.3 Accelerated aging. The tensile strength of the tube and cover after the accelerated aging process described in 4.6.6 shall not be less than 80 percent of the tensile strength indicated before the aging process, and ultimate elongation shall be not less than 50 percent of the values specified in Table I.

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3.5.4 Ozone resistance. The cover compound of the hose shall show no visible cracking under 2X magnification after exposure to ozone as specified in 4.6.8.

3.6 Identification marking. Each length of hose shall be marked on the cover in two places, approximately 4 feet from the ends. The marking shall be in letters at least 0.125 inch high and shall include: (a) the manufacturer's name or trademark, (b) the maximum working pressure (e.g., working pressure 400 psi), and (c) the quarter of the calendar year and the year of manufacture.

3.7 Dimensions.

3.7.1 Hose diameter. The inside diameter of the hose shall be 2 inches or 3 inches (±0.062), as specified (see 6.2).

3.7.2 Length. Unless otherwise specified (see 6.2), the hose length shall be 50 feet. The length of the hose shall be measured from the ends of the hose, not from the ends of the couplings.

3.7.2.1 Tolerance. A tolerance of ±1 percent shall be permitted in the length of the hose.

3.8 Workmanship. The finished hose assembly shall be free of any defect that may affect its appearance and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection requirements shall be classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one sample hose length when a first article sample is required (see 3.1 and 6.2). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be a standard production item from the contractor's current inventory provided the unit meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining units to be furnished under the contract.

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4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on the sample hose lengths selected in accordance with 4.4. This inspection shall include the examination of 4.5 and the tests of 4.6.

4.3 Inspection lot. All hose and hose assemblies of the same type and size offered to the Government at one time shall be considered a lot for the purpose of inspection.

4.4 Sampling. A random sample of hose lengths shall be selected from each lot in accordance with MIL-STD-105.

4.4.1 Sampling for examination. Examination of the hose shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.2 Sampling for tests. Tests of the hose shall be based on inspection level S-3 with an AQL of 4.0 percent defective. A representative sample 3 feet in length shall be cut from lengths selected from each lot. The contractor shall reattach couplings to the remaining length from which the 3 foot length was cut. After the couplings have been reattached, the remaining length of the hose shall be used to conduct the hydrostatic proof pressure test specified in 4.6.1. When the remaining length has passed the hydrostatic proof test, it shall be accepted as a full length of hose. At the completion of the burst test, tests for aging, resistance to ozone, resistance to water, resistance to oil, adhesion, tensile strength and elongation shall be conducted on the 3-foot sample used in the burst test.

4.5 Examination. Each of the sample hose lengths selected in accordance with 4.4.1 shall be visually and dimensionally examined to determine conformance with all the requirements of this specification not involving tests. The classification of defects in Fed. Std. No. 162 shall be used to determine and evaluate defects through visual inspection.

4.6 Tests. Sample hose sections selected in accordance with 4.4.2 shall be subjected to the applicable test specified in Table II and 4.6.1 through 4.6.8.

TABLE II. Test method references.

Test	Requirement paragraph	Method in Fed Test Method Std. No. 601 or ASTM	Reference paragraph
Hose size (inside diameter)	3.7.1	2221	
Hose length	3.7.2	2411	
Tube and cover thickness	3.2.1 and 3.2.3	2011	
Proof pressure	3.5 and Table I	ASTM D380	4.6.1
Burst pressure	3.5 and Table I	ASTM D380	4.6.2
Tensile strength and elongation	3.5 and Table I	ASTM D412	4.6.3
Swelling	3.5.1.1	6211	4.6.4
Accelerated aging	3.5.3	ASTM D573	4.6.5
Adhesion	3.5 and Table I	ASTM D413	4.6.6
Oil immersion	3.5.2	6111	4.6.7
Ozone	3.5.4	ASTM D1149	4.6.8

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4.6.1 Hydrostatic proof test, coupled assemblies. Samples selected in accordance with 4.4.2 after couplings have been reattached shall be subjected to the hydrostatic proof test pressure specified in Table I in accordance with ASTM D380. Water shall be used as the test media. Leakage or other evidence of defects shall be cause for rejection.

4.6.2 Burst test. Hose samples selected in accordance with 4.4.2 shall be subjected to the hydrostatic straight bursting test of ASTM D380, after which they shall be used to perform the other tests shown in Table II. Water shall be used at the test media. Any suitable coupling other than those specified in 3.4 may be used for this test. Failure of the hose at a pressure below the minimum burst pressure specified in Table I shall be cause for rejection of the lot.

4.6.3 Initial tensile properties. Initial tensile strength and ultimate elongation shall be determined by the procedure given in ASTM D412, using specimens cut with die C.

4.6.4 Volume change after immersion. The change in volume of samples of the tube, after immersion for 22 $\pm$1/4 hours in distilled water shall be determined in accordance with method 6211 of Fed. Test Method Std. No. 601. The change in volume shall be reported as a percent of the original volume.

4.6.5 Accelerated aging test. Hose samples selected in accordance with 4.4.2 shall be subjected to accelerated aging in accordance with ASTM D573, except that the time for aging shall be 70 hours at a temperature of 212° Fahrenheit (F). After aging, the tensile strength shall be not less than 80 percent of the values specified in Table I, and the ultimate elongation shall be not less than 50 percent of the values specified in Table I. The die for cutting the specimen shall be die C of ASTM D412.

4.6.6 Adhesion. From hose assemblies selected under 4.4.2, prepare ring specimens as described in ASTM D380. The adhesion shall be determined in accordance with the machine method of ASTM D413. The rate of travel of the power-actuated grip shall be 1 inch per minute. Nonconformance to the requirements of Table I shall constitute failure of this test.

4.6.7 Oil immersion. The tensile strength of the tube and cover, after immersion in oil No.1 of ASTM D471, shall be determined in accordance with method 6111 of Fed. Test Method Std. No. 601, using specimens cut with die C of ASTM D412. Tensile strength shall be based on the original cross-sectional area of the specimen.

4.6.8 Ozone resistance test. Specimens of the cover, prepared as described in Procedure A of ASTM D518, shall be mounted in a 20 percent elongated position and tested in accordance with ASTM D1149. After conditioning for 24 hours in an ozone-free atmosphere, the mounted specimen shall be exposed for 72 hours at 104°F $\pm$2°F to an atmosphere containing 50 parts per hundred million of ozone. Visible cracking under 2X magnification at the end of such test shall be cause for rejection.

4.7 Packaging inspection. The preservation-packaging, packing, and marking of the hose shall be inspected to verify conformance to the requirements of Section 5 of this specification.

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5. PACKAGING

5.1 Preservation-packaging, packing, and marking. The hose shall be preserved, packaged, packed, and marked in accordance with MIL-P-775. The level of preservation-packaging and level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The hose covered by this specification is intended for pile driving, grading, surface stripping, hydraulic mining, air compressor use, and other applications requiring high pressure water service and air service.

6.2 Ordering data. Acquisition documents shall specify the following:

(a) Title, number, and date of this specification.
(b) If first article is required for inspection and approval (see 3.1, 4.2.1, and 6.5).
(c) Size hose required (see 3.7.1).
(d) If hose length shall be other than 50 feet (see 3.7.2).
(e) Level of preservation-packaging and level of packing required (see 5.1).

6.3 Recommended working pressure. Recommended working pressure for the hose covered by this specification is 400 psi.

6.4 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.5 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete hose length or hose assembly or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all acquisition instruments regarding arrangement for examinations, tests, and approval of the first article.

6.6 Definitive specification part number. The specification part number is a definitive part number which will be formulated to identify each item covered by this specification. The part number will be formulated by selecting from the requirement options available in this specification as follows:

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Hose assembly

Definitive military specification part number	M28523	X	XX	XX
Military specification number				
Hose size code letter (see 6.6.1)				
Length in feet				
Length in inches				

6.6.1 Hose size. Hose size is designated by one code letter (see Table III).

TABLE III. Hose size code letter.

Hose size code letter	A	B
Hose inside diameter (inches)	2	3

Custodian:

Navy - YD

Review activity:

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Preparing activity:

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